MIL-L-52292B(ME)
8 June 1979
SUPERSEDING
MIL-L-52292A(ME)
28 October 1971

MILITARY SPECIFICATION

LIGHT ASSEMBLY, MARKER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a light-weight, self-contained, light assembly which emits visible light of selective color in variable intensity. The light assembly can be operated from an internal 10-volt nickel cadmium battery or from an external power source. The illumination is 9 candlepower, and the distribution is 360 degrees in the horizontal and 180 degrees in the vertical plane. The light assembly is a component of the light set, marker, landing and drop zone.

FSC 6230

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-636	- Boxes, Shipping, Fiberboard.
QQ-S-781	- Strapping, Steel and Seals.

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-V-173	- Varnish, Moisture-and Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-T-704	- Treatment and Painting of Materiel.

STANDARDS

FEDERAL

FED. STD. No. 595	- Colors.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables For Inspection by Attributes.
MIL-STD-129	- Marking For Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-202	- Test Methods For Electronic and Electrical Component Parts.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements For Equipment.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

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DRAWINGS

ME

TA13201E3050 - Light Assembly, Marker.

(Copies of specifications, standards, and drawings required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

SOUTHERN CALIFORNIA AIR POLLUTION CONTROL DISTRICT

Rule 102 - Photochemically Reactive Solvents

(Application for copies should be addressed to the Southern California Air Pollution Control District, 434 S. San Pedro Street, Los Angeles, CA 90013.)

3. REQUIREMENTS

3.1 Description. The light assembly shall be in accordance with Top Assembly TA13201E3050 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the light assembly. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer of his designated representative.

3.2 First article (preproduction model). The contractor shall furnish one light assembly as specified (see 6.2) for examination and testing within the time frame specified (see 6.2) to prove prior to starting production that his production methods will produce light assemblies that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

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A fixture shall be used for mounting the lamp with the electrodes in a

3.3 Operational.

3.3.1 Internal power. The light assembly, with the battery fully charged and the operation mode switch in either the "FLASH" or "ON" positic shall give off a beam of light as specified in 1.1.

3.3.2 External power. The light assembly connected to a 100 volt, 60 or 400 Hertz power supply shall operate in each of three modes of operatio "ON", "FLASH" and "CHARGE".

3.4 Beam characteristics. The initial candlepower of the light shall b not less than 9 with the light assembly operating from either the external or internal power supply and with the clear lens installed. When tested as specified in 4.5.3 the candle power shall not be less than 6 or 6.5 as applicable.

3.5 Temperature and humidity. The light assembly shall not malfunctio or be damaged by operation at ambient temperatures from plus 125° to minus 25° F. And at all relative humidities possible at these temperatures. The light assembly shall not be damaged by storage at ambient temperatures from plus 160° to minus 50° F and at all relative humidities possible at these temperatures.

3.6 Flash rate. The electrical circuit that controls the flash rate o the lamp shall provide a flash rate of 30 cycles per minute. The "ON" and "OFF" periods shall be equal.

3.7 Waterproofing. When tested as specified in 4.5.4, the light assembly shall not be permanently damaged (see 6.4) and shall operate as specified herein within 3 minutes after removal from the water.

3.8 Operational performance. The light assembly shall operate as specified in 3.3 for a continuous period of 10 discharge and charge cycles when the light is operated from its internal battery and for 10 hours continuous operation when the light is operated using external power source.

3.9 Ruggedness. The light assembly shall withstand without damage (se 6.4), the usage encountered in military operations such as air drop, and transportation over rough terrain in military vehicles.

3.10 Electromagnetic interference. The electromagnetic interference emission characteristics of the light assembly shall conform to MIL-STD-46 Class No. IIA.

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3.11 Treatment and painting. The light assembly components normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, Type B. Color shall be No. 34087, olive, as specified in FED. STD. No. 595.

3.12 Fungus and moisture resistance. The electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, Composition I or II, as applicable (see 3.12.1), with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish.

(a) Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.12.1 Composition. Composition II shall be used only in the case when local air pollution regulations governing application of varnish precludes the use of Composition I. When Composition II is used, the contractor shall provide evidence to the Government that the use of Composition II is required, and shall certify that the Composition II material complies with Southern California Air Pollution Control District, Rule 102, Photochemically Reactive Solvents.

3.13 Identification marking. The identification marking shall be in accordance with MIL-STD-130 and as shown on the applicable drawings.

3.14 Workmanship. Joints and seams shall fit tightly, edges shall be smooth, and welded and soldered surfaces shall be cleaned. Optical surfaces shall be free of scratches or blemishes.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities

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suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction light assembly shall be examined in accordance with Table I. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. Following successful completion of the examination, the preproduction light assembly shall be subjected to the tests marked "X" in Column 1 of Table II. Tests shall be conducted in the order listed. Failure of any tests shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105. Inspection Level S-2 shall be used for lot sizes of 2 to 65. For lot sizes of 66 and larger, Inspection Level S-3 shall be used.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined in accordance with Table I and compared with the preproduction model. AQL shall be 2.5 percent for major defects and 6.5 percent for minor defects.

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Table I. Defects

Defect		Requirement Paragraph
Major		
101.	Critical dimension not as shown.	3.1
102.	Absence of parts or components.	3.1
103.	Misalinement of parts or components.	3.1
104.	Materials or components not as specified.	3.1
105.	Dials not legible.	3.1
106.	Wiring and connections not as specified.	3.1
107.	Workmanship not as specified.	3.14
Minor		
201.	Treatment and paintings not as specified.	3.11, 3.12
202.	Identification marking missing or incorrect.	3.13

4.4.3 Tests.

4.4.3.1 Individual. Each light assembly shall be subjected to the test marked "X" in Column 2 of Table II. Failure of the test shall be cause for rejection.

4.4.3.2 Samples. Samples selected in accordance with 4.4.1 and which have passed the individual test specified in 4.4.3.1 shall be subjected to tests marked "X" in Column 3 of Table II. The operational reliability test shall be limited to five charge and discharge cycles. AQL shall be 2.5 percent defective for major defects and 6.5 percent defective for minor defects.

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Table II. Test Schedule

Prepro. model	Production		Test	Inspection paragraph	Requirement paragraph
	Indiv.	Sample			
1	2	3	4	5	6
X	X	-	Operation	4.5.1.1 4.5.1.2	3.3, 3.6
X	-	X	Beam characteristics	4.5.2	3.4
X	-	-	Temperature and humidity	4.5.3	3.5
X	-	-	Waterproofing	4.5.4	3.7
X	-	X	Operational performance	4.5.5	3.8
X	-	-	Ruggedness	4.5.6	3.9
X	-	-	Electromagnetic interference	4.5.7	3.10

4.5 Test procedures.

4.5.1 Operation.

4.5.1.1 Internal power. Operate the light assembly, with a fully-charged battery installed, for 30 minutes with the mode switch in the continuous "ON" position and 30 minutes with the mode switch in the "FLASH" position. Malfunction of the light in either mode of operation or failure to conform to the flash rate specified in 3.6 shall constitute failure of this test.

4.5.1.2 External power. Connect the light assembly to a 100 volt, 60 Hertz power supply and operate with the mode switch in the continuous "on" position for 30 minutes and then with the mode switch in the "flash" position for 30 minutes. After operation in the continuous "on" and "flash" positions, rotate the mode switch to the "charge" position and increase the input voltage to 110 volts. Charge the batteries for 30 minutes. Repeat the test procedure using 400 Hertz power supply. Failure of the light assembly to operate in either mode, or failure of the battery in the light assembly to recharge shall constitute failure of this test.

4.5.2 Beam characteristics. Equip the light assembly with a clear lens and operate with the mode switch in the continuous "on" position. Conduct the test in two phases: (1) Operate the light assembly from its own battery

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(2) operate each light assembly from a 100 volt, 60 Hertz ac source. Measure the candlepower with a viscor filtered photometer calibrated against a National Bureau of Standards (NBS) standard lamp operating at a color temperature of 2879° Kelvin. Readings shall be taken perpendicular to the longitudinal axis of the lamp filament and at 5 degrees from the perpendicular. Two readings on each light assembly in each test phase shall be taken. Repeat the test procedure using 400 Hertz power supply. An average reading of less than 9 candlepower shall constitute failure of this test.

4.5.3 Temperature and humidity. The temperature in the test chamber shall be maintained within 4° F of test temperature throughout the test.

4.5.3.1 Low-temperature storage. Place the light assembly in a test chamber and expose it to a temperature of minus 50° F for 12 hours prior to starting the tests. Outside the chamber, operate each light assembly from its internal battery for 1 hour with the mode switch in "flash" position. Measure the caldlepower of each light assembly at the end of 1 hour. Conduct the same test on the light assembly except that auxiliary power (100 volt, 60 Hertz) shall be used. Examine for damage at the conclusion of the test. A reading on any light assembly of less than 6 candlepower when operating on its internal power source or of less than 9 candlepower when operating from an external power source, or evidence of damage shall constitute failure of this test.

4.5.3.2 Low-temperature operation. Place the light assembly in a test chamber and expose it to a temperature of minus 25° F for 12 hours prior to starting the tests. Inside the chamber while maintaining the minus 25° F temperature, operate each light assembly from its internal battery for 1 hour with the mode switch in "flash" position. Measure the candlepower of each light assembly at the end of 1 hour. Conduct the same test on the light assembly except that auxiliary power (100 volt, 60 Hertz) shall be used. Examine for damage at the conclusion of the test. A reading on any light assembly of less than 6 candlepower when operating on its internal power source or of less than 9 candlepower when operating from an external power source, or evidence of damage shall constitute failure of this test.

4.5.3.3 High-temperature and humidity storage. Place the light assembly in a test chamber and expose it for 12 hours to an ambient temperature of plus 160° F at maximum relative humidity prior to starting the tests. Outside the chamber, operate each light assembly for 2 hours from its internal

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battery with the mode switch in the "flash" position. Read the candlepower of each light assembly at the end of 2 hours. The same test shall be conducted on the light assembly except that the auxiliary power (100 volt, 60 Hertz) shall be used. Examine for damage at the conclusion of the test. A reading of less than 6.5 candlepower at the end of 2 hours when operating on its internal power source or of less than 9 candlepower when operating from an external source, or evidence of damage shall constitute failure of this test.

4.5.3.4 High-temperature and humidity operation. Place the light assembly in a test chamber and expose it for 12 hours to an ambient temperature of plus 125° F at maximum relative humidity prior to starting the tests. Inside the chamber while maintaining the plus 125° F temperature, operate each light assembly for 2 hours from its internal battery with the mode switch in "flash" position. Read the candlepower of each light assembly at the end of 2 hours. The same test shall be conducted on the light assembl except that the auxiliary power (100 volt, 60 Hertz) shall be used. Examine for damage at the conclusion of the test. A reading of less than 6.5 candlepower at the end of 2 hours when operating on its internal power source or of less than 9 candlepower when operating from an external source, or evidence of damage shall constitute failure of this test.

4.5.4 Waterproofing. Immerse the light assembly in 3 feet of salt water of specific gravity of 1.04 for 1 hour and remove. Operate the light assembly within 3 minutes after removal from the salt water. Then test the light assembly for operation in accordance with 4.5.1 except that operation in the "flash" position only is required. Nonconformance to 3.7 shall constitute failure of this test.

4.5.5 Operational performance. Operate the light assembly from its inte battery with the mode switch in the "flash" position. Recharge the battery by use of a 120 volt, ac supply connected to the light assembly. Repeat thi test 10 times. The light assembly source internal battery is considered discharged when the voltage at the lamp drops below 6.5. The battery is considered fully charged when the no-load voltage at the lamp is 12. After completion of this test, the light assembly with the mode switch in "flash" position shall be operated continuously for 10 hours using a 100 volt, ac, power source. Inability of the light assembly to operate through the number of cycles specified or to operate for an ac source continuously for 10 hours shall constitute failure of this test.

4.5.6 Ruggedness.

4.5.6.1 Shock. The light assembly shall be tested as specified in MIL-STD-202, Method 213, test condition letter optional. Upon completion of the shock test, perform the tests specified in 4.5.1.1 and 4.5.1.2. Failure to operate under any specified input conditions or evidence of damage shall constitute failure of this test.

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4.5.6.2 Vibration. The light assembly shall be tested as specified in MIL-STD-202, Method 201. Upon completion of the vibration test, perform the tests specified in 4.5.1.1 and 4.5.1.2. Failure to operate under any specified input conditions or evidence of damage shall constitute failure of this test.

4.5.7 Electromagnetic interference. The preproduction model, equipped for the reduction of electromagnetic interference in accordance with 3.10 shall be tested in accordance with MIL-STD-461. The contractor shall furnish the contracting officer the test plan and test report required by MIL-STD-461 at the time interval specified by contracting officer. Upon approval of the report by the contracting officer and provided all other requirements of the specification are met, the preproduction model shall shall be used as a model for all production light assemblies. Disapproval of the report shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective for major defects and 6.5 percent defective for minor defects.

Major

108. Materials, methods, or containers not as specified for Level A or B. Each incorrect material, method, or container shall be considered one defect.
109. Lights not individually preserved as specified for Level A or B.
110. Gross weight exceeds 200 pounds for Level A.
111. Strapping not as specified.
112. Gross weight or size exceeds the box limitations for Level B.

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Minor

203. Marking missing, illegible, incorrect, or incomplete.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A, Level B or Commercial as specified (see 6.2).

5.1.1 Level A. Each light shall be individually preserved in accordance with MIL-P-116, Method IC-2, using a close-fitting commercial fiberboard box or a box conforming to PPP-B-566 or PPP-B-636, Type CF, Class Domestic, Variety SW, Grade 125. The contents shall be cushioned within the box in a manner to prevent movement or damage.

5.1.2 Level B. Each light shall be individually preserved in accordance with MIL-P-116, Method III, using a close-fitting box conforming to PPP-B-636, W6c, style optional. The contents shall be cushioned within the box in a manner to prevent movement or damage.

5.1.3 Commercial. The lights shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, Level B

5.3 Marking.

5.3.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

5.3.2 Commercial packaging. Marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The light assembly is intended for use in marking landing strips, drop zones, and aircraft parking areas.

6.2 Ordering data. Procurement documents should specify the following:

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(a) Title, number, and date of this specification.
(b) Time frame required for submission of preproduction model.
(c) Time interval for submission of test plan and test report (see 4.5.7).
(d) Degree of preservation and degree of packing required (see 5.1 and 5.2).
(e) When Finish A strapping is required (see 5.2.1).

6.3 Preproduction model. Any changes or deviations of production light assemblies from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish light assemblies conforming to this specification.

6.4 Damage. Damage is defined as a permanent deformation, loosening, breakage, corrosion, or change of fit or any component or part, or any condition resulting in nonconformance to any requirement in this specification.

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the right to perform any of the inspections set forth in the speci-

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